Sora | AI Work Companion

Technical report - version 1.7 - 26 October 2026

A desktop companion that uses visual evidence to guide considerate work reminders and adapt its next intervention to the observed outcome.

Release status: runnable local prototype. AWS deployment tooling is prepared; a live AWS deployment and representative real-video accuracy have not yet been verified. These are required final team tasks, not completed results.

Problem, users and intended value

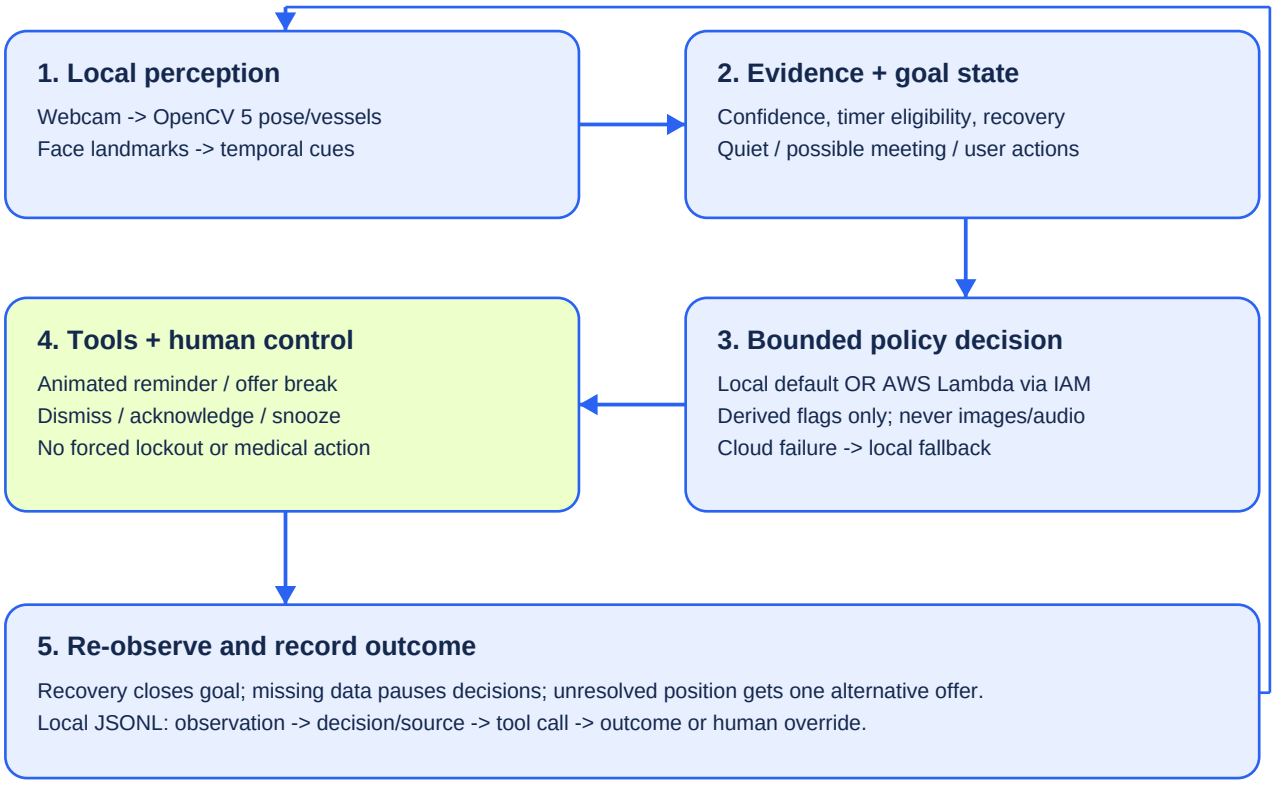
Students, programmers and other desk-based computer users can lose track of posture changes and time spent working. Generic alarms lack context: they may interrupt a call, repeat after dismissal, or assume compliance without observing an outcome. Sora aims to make reminders contextual, inspectable and easy to control. No clinical or productivity benefit has been measured.

Implemented scope

Capability	Behavior and boundary
Posture	Personal head/shoulder reference; sustained lower-head observations over a configurable window. Not spine-angle measurement.
Water	Timer plus likely cup/bottle approach-contact-retreat events; manual confirmation and undo. Experimental straw geometry is off by default.
Distance / fatigue cues	Relative face size, prolonged eye closure and repeated possible yawns. Not centimeters or a fatigue diagnosis.
Breaks / session limit	Observed-presence timers for movement, screen breaks and session limit; elapsed unpaused time for water. No sleep or gaze verification.
Companion interaction	Figma-derived interface, six original GIFs, one English caption and three icon actions on a transparent overlay.
Context and agent	Optional possible-meeting suppression; bounded outcome-aware policy with local traces and optional AWS execution.

Scope is personal assistance, not workplace surveillance. Start/Calibrate activates the webcam; Pause, End and Quit release it.

Architecture and agent workflow



Substantive OpenCV 5: person detection and pose inference produce head/shoulder measurements. NanoDet with alternative views, YOLOX fallback and ViT vessel tracking support temporal drinking-motion estimates. OpenCV also processes brightness, overlays and experimental straight-straw line geometry. MediaPipe face landmarks complement this pipeline.

Closed loop: an eligible posture observation first invokes the reminder tool. A stable return toward the reference resolves the goal. If the position remains unresolved, the agent can offer a break once after 120 seconds (8 seconds in Demo). Missing evidence pauses intervention. Dismiss blocks follow-ups; snooze defers them. Repeated UI polling must not create extra tool calls.

Policy and cloud: this is an explicit deterministic policy agent, not an LLM. In cloud mode, IAM-authenticated Lambda evaluates the bounded decision contract; the desktop validates the result and executes the tool. An asynchronous request keeps the camera responsive, rejects stale decisions and falls back locally after three seconds. Quiet mode and uncertain evidence override cloud actions immediately.

Minimal cloud footprint: one 128 MB Lambda, one execution role and one seven-day log group. No public endpoint or camera upload. The cloud component is ready to deploy, but no AWS execution is claimed in this report.

Validation and reproducibility

Evidence	Result	What it establishes
Automated tests	75 passed	Timers, session/API behavior, model adapter use, agent transitions, human controls, synthetic cues, metrics and cloud contract/failure handling.
Synthetic agent replay	6 / 6 passed	Recovery, unresolved posture, tracking loss, meeting deferral, dismissal and water confirmation. Not camera accuracy.
Browser checks	Passed locally	Figma Home, Agent activity, opt-in meeting toggle and settings save; no JavaScript errors. Camera disabled.
Real-video evaluation	Tool ready; data pending	Evaluator runs actual model inference and exports event precision/recall, duplicates, misses, coverage and offline throughput.
AWS deployment	Not yet verified	Template and handler contract tested locally; account deployment and real invocation receipt still required.

Re-run: `python -B -m unittest test_ui test_timers test_visual_features test_agent test_delivery -q`

Reproduce policy traces: `python replay_agent.py`. The outputs explicitly identify synthetic policy input. Local live traces are saved under sessions/*.agent.jsonl.

Real-video protocol: use consented recordings with three initial reference seconds, cup/bottle/straw cycles and negative actions. Human-reviewed start/end annotations form the ground truth. One event can match only one interval; extra events count as false positives. Keep evaluation clips separate from tuning clips. Report sample size, vessel mix, lighting and failures alongside metrics.

Known failure cases

Mouth occlusion, transparent vessels, head rotation, glasses and weak lighting can cause missed or incorrect visual estimates. Straw geometry may confuse other straight edges and is disabled by default. Sustained speech can resemble a yawn. Windows microphone metadata can miss muted/packaged calls or mistake other microphone use for a meeting. A camera move invalidates the calibration. These limitations require real-device testing; software tests alone do not resolve them.

Operation, privacy and final delivery

Setup: extract the complete archive to a fresh folder on Windows, run Setup Sora UI.bat, then Start Sora UI Demo.bat. Python 3.14 and pinned OpenCV 5 dependencies are specified. Models, model provenance, Figma assets and the original GIFs are included. Use the normal launcher for real reminder intervals.

Human control: users can disable individual features, dismiss/snooze reminders, correct water events, enter Quiet mode, override meeting suppression and stop the session. The agent never locks the screen, diagnoses a condition or claims a user slept. Startup is opt-in and does not activate the camera.

Data: webcam frames remain in local memory. Session summaries and traces are local files; cloud mode sends only a goal, stage and five boolean flags. Lambda logs only policy action. Optional meeting detection reads microphone-use metadata and running executable names, never audio or window contents. Cloud credentials use the AWS SDK credential chain and are excluded from the release.

AWS steps: inspect cloud/template.json, authenticate an account locally, run cloud/deploy.py with the intended account and region, and retain the real deployment receipt. Demonstrate an AWS-sourced trace and a network-failure fallback. Review charges in the chosen account; cleanup removes the named stack. Full commands are in cloud/README.md.

Final team checklist

Needed before final submission	Prepared material
Verify AWS deployment	CloudFormation template, deployment/verification/cleanup script and runtime IAM policy generator.
Measure actual camera behavior	Video evaluator, annotation format, test protocol and failure-case checklist.
Record team and application demo	Timed 4:30 script; show architecture, actual results and human controls.
Provide judge-accessible distribution	Complete application archive, technical report, source/tests and draft Devpost story. Add actual repository/archive and video URLs.

The competition requires substantive OpenCV 5 work and a meaningful AWS component. For the Agentic Vision path, demonstrate that visual evidence changes a later action and provide trace/evaluation evidence. Eligibility and competitive performance remain subject to judging.

Official sources: Competition requirements; official award rubric; AWS Lambda Invoke documentation. Checked 26 October 2026.

Use the earlier displayed deadline: 26 October 2026, 23:45 PDT / 27 October, 14:45 WITA. Verify the actual submission form; the overview body displays a later time.